

schematics for electrical errors and warnings. You will be surprised to see the number of warnings and errors reported by the tool for the schematic that you prepared and thought was perfect.

Step 2: Create a PCB layout of the schematic

You may choose to use either through-hole components or surface-mount components (see Fig. 1), or a combination of both. This step remains the same for both types of components, except that some extra precaution needs to be exercised for through-hole components. Many software tools like Eagle allow you to import all the components from the schematic created by you to the board layout view while concurrently validating the board layout view (*aka* board view) electrically against the schematic. In this view, the board boundary is displayed, with all the components placed outside it, to begin with.

You can then start placing components within the board boundary by dragging them, one at a time. All the components show lines called 'airwires' to display the electrical connectivity between their pins and pins of other components they connect with. This ensures that you do not miss out on any electrical connection inadvertently. By the end of this step no 'airwires' should be visible and the board layout should be complete.

Eagle also provides the design rules check (DRC) feature, which checks the board against a set of predefined design rules. You will find similar features in other tools as well.

Tips, pitfalls and workarounds. If you use through-hole components, you need to drill holes for the component leads (step 8). A popular choice for drilling holes is a mini-push-type hand-drill machine as shown in



Fig. 4: Hand drill and bit

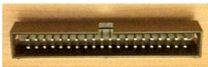


Fig. 5: 40-pin DIL header

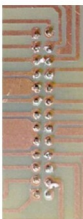


Fig. 6: Messed up PCB pads

Fig. 4. The bit normally used to make through-holes in the PCB has 1 mm dia. These drills are fine for 40-pin DIL header (Fig. 5) and drilling holes for components like resistors, capacitors and transistors, where a degree of misalignment is acceptable because you can handle it at the time of mount-

ing components. However, it is very difficult to drill multiple holes perfectly in a straight line with exactly the same spacing (usually 2.54mm), like dual-line pins of an IC (see Fig. 13) or a header block (see Fig. 5), without any alignment error. And it is highly likely that you will end up in the situation shown in Fig. 6.

You will notice that the drill hole almost consumes the entire component pad, leaving no copper on the pad to hold any solder to bind the track connected to the pad with the component lead. This happens for a number of reasons:

1. The quality and precision with which a hole can be drilled by the mini push-drill (shown in Fig. 4) leaves a lot to be desired. Typically, for PCB application the minimum prescribed speed is 25,000rpm, which may go up to 200,000rpm for multilayer microvias, whereas manually operated hobby mini push-drills operate at a very low rpm and wobble

2. Most PCB laminates are made of fiberglass or glass epoxy, which are abrasive materials and wear away steel bits really fast. One solution is to use solid tungsten carbide bits, which last about ten times longer than steel bits. Unfortunately, these bits can't be used with mini-push-drills as they are very brittle and break with the slightest change in the angle of contact

3. It is virtually impossible to keep the drill absolutely perpendicular to the PCB at all times while pushing the plunger of the drill axially up and down

Commercial PCB manufacturing houses have appropriate manufacturing equipment that address the above three aspects. Eagle and most other software solutions use default parameters like pad diameter and shape (see Eagle default pad size and shape for jumpers in Fig. 7), which can be realised by commercial PCB making houses. For home brewing PCBs you need to work around these limitations. One way to do this is to use a bigger component pad size, but here is a trade-off: If you use very big pad size, you may end up having solder bridges across the component pads. If the pad size is too small, the drill hole may occupy the entire pad leaving no copper to solder on.

What parameters you use is also a function of the component type. Here we consider three types of components: discrete two-legged components like resistors, capacitors, transistors and jumpers; connectors like header blocks that have single-in-line or dual-in-line pins; and integrated circuits (ICs). These tips can also be used for other components with care and caution.

For discrete two-legged components, examine Fig. 7 for the default pad size and shape used by Eagle for jumpers. Fig.